

ADVANCED EXCEL

Data Analysis Case Study

Conditional Logic, Lookup Functions, Aggregation & Dashboarding in Microsoft Excel

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Project Repository:

github.com/princemaheta2627-glitch/excel_case_study_2

Category: Data Analytics / Data Science Portfolio Project

Tool Used: Microsoft Excel (Formulas, PivotLogic, Charts, VLOOKUP, Data Validation)

1. Executive Summary

This project is a two-part Advanced Excel case study demonstrating core data analysis skills that translate directly into a data science workflow: data cleaning and structuring, rule-based classification, aggregation and summary statistics, lookup-driven data retrieval, and visual storytelling through charts. The first part analyzes a student examination dataset to classify performance and summarize results by region. The second part works with real-world financial time-series data (NSE Nifty 50 index, March-June 2021) to build a dynamic lookup tool and trend visualizations.

While the toolset is Excel rather than Python or SQL, the underlying analytical thinking — defining business rules, aggregating data by category, building repeatable lookup logic, and communicating insights visually — is identical to what is expected in a professional data analytics/data science role. This report documents the objective, dataset, methodology, formulas used, and key insights for each part of the project.

2. Project Objective

- Convert raw, ungrouped data into a structured, rule-based classification system using nested logical functions.
- Summarize and compare categorical groups (regions/states) using conditional aggregation functions.
- Build a dynamic, searchable lookup system for time-series data using VLOOKUP.
- Visualize trends and relationships in financial data through appropriately chosen charts.
- Demonstrate Excel as a legitimate first-line data analysis and reporting tool, prior to scaling into Python/SQL/BI tools.

3. Tools & Techniques Used

Category	Techniques Applied
Software	Microsoft Excel (formulas, tables, charts, data validation)
Logical Functions	Nested IF() for multi-tier grade classification
Conditional Aggregation	COUNTIF(), SUMIF(), AVERAGEIF()
Ranking	RANK() for comparative performance scoring
Text Processing	CONCATENATE() to build readable summary strings
Lookup / Retrieval	VLOOKUP() with an interactive date-based lookup table
Data Structuring	Excel Tables for structured, auto-expanding ranges
Visualization	Clustered column charts, combination (column + line) charts

4. Part A — Student Examination Performance Analysis

4.1 Dataset Description

4.2 Grade Classification Logic

Condition	Classification
Marks < 35	FAIL
$35 \leq \text{Marks} < 50$	PASS
$50 \leq \text{Marks} < 60$	2nd Class
$60 \leq \text{Marks} < 75$	1st Class
Marks ≥ 75	DISTINCTION

A companion CONCATENATE() formula then generated a single readable result string per student, e.g. combining first name, last name, state, marks and result status into one field — useful for building report-ready output or merge-fields for report cards.

Figure 1: Student-level dataset with nested-IF grade classification and concatenated summary column.

4.3 State-Level Summary (Conditional Aggregation)

To compare performance across states, COUNTIF(), SUMIF() and AVERAGEIF() were used to build a summary table, and RANK() was applied to rank the four states by average marks.

For the following dataset:

1 Mark the student as **FAIL** if the marks are less than 35 and **PASS** - if the marks are 35 and 75, and **DISTINCTION** - if the marks are above 75, in the column E, by using IF

2 Fill in the following table:

City	CountIF()	SumIF()	Average Marks	Rank
Delhi	6	417	69.5	1
Gujarat	6	322	53.66667	3
Bihar	4	197	49.25	4
Orissa	4	232	58	2

4 Convert the data given in the Dataset in the form a table, using the formulae/fur

5 Concatenate the data in Column G as "LastName FirstName Marks Status of Pas

Roll No	First Name	Last Name	Marks	Result	State
1	Masum	Chopra	59	2nd Class	Delhi
2	Perna	Agarwal	20	FAIL	Gujarat
3	Rahil	Saini	56	2nd Class	Bihar
4	Pooja	Patel	75	DISTINCTION	Orissa
5	Mohak	Chaudhary	68	1st CLASS	Delhi
6	Mini	Desai	95	DISTINCTION	Gujarat
7	Omi	Patel	38	PASS	Delhi
8	Parshwa	Kumar	63	1st CLASS	Gujarat
9	Rohit	Chopra	84	DISTINCTION	Delhi
10	Kiran	Singora	45	PASS	Gujarat
11	Rahil	Chopra	22	FAIL	Bihar
12	Sree	Agarwal	69	1st CLASS	Orissa
13	Ram	Saini	78	DISTINCTION	Delhi
14	Piku	Patel	54	2nd Class	Gujarat
15	Naren	Chaudhary	85	DISTINCTION	Bihar
16	Neeraj	Desai	28	FAIL	Orissa
17	Maya	Patel	90	DISTINCTION	Delhi
18	Malvika	Kumar	45	PASS	Gujarat
19	Monika	Chopra	34	FAIL	Bihar
20	Shobhit	Singora	60	1st CLASS	Orissa

Figure 2: State-wise student count, total marks, average marks and performance rank.

4.4 Key Insights — Part A

- Delhi recorded the highest average marks (69.5) and ranked #1 among the four states.
- Orissa followed in 2nd place with an average of 58 marks.
- Gujarat placed 3rd (53.7 average) despite having the same student count as Delhi (6 students), indicating more inconsistent performance within the group.
- Bihar had both the fewest students (4) and the lowest average (49.25), ranking last.
- The spread between the top and bottom state averages is roughly 20 marks — a meaningful gap that would justify targeted academic support for Bihar and Gujarat in a real institutional setting.

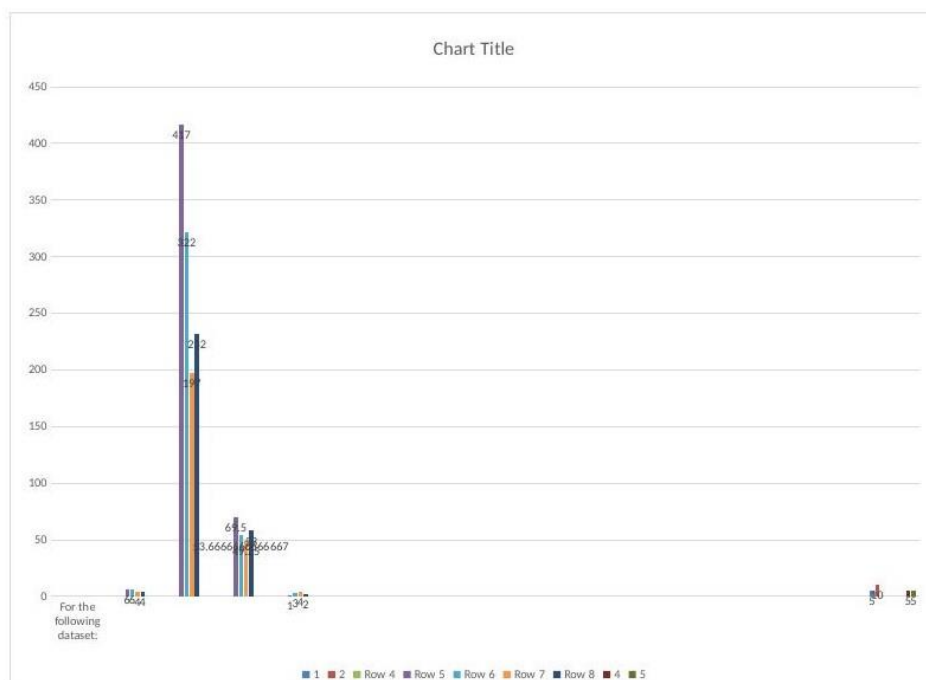


Figure 3: Chart visualization comparing question-wise scoring and state performance metrics.

5. Part B — NSE Nifty 50 Index Trend Analysis

5.1 Dataset Description

A real-world financial time-series dataset containing daily Open, High, Low, Close, Shares Traded and Turnover (Rs. Cr) values for the Nifty 50 index, spanning March to June 2021 (~80 trading days). This mirrors the type of OHLC (Open-High-Low-Close) market data commonly analyzed in finance and quantitative analytics roles.

5.2 Dynamic VLOOKUP Lookup Table

An interactive lookup system was built so that selecting any date from the dataset instantly returns that day's Open, High and Low values — a lightweight version of a lookup-driven dashboard control.

Formulas used:

=VLOOKUP(R9,A6:G88,2,FALSE) → returns Open price for the selected date

=VLOOKUP(R9,A6:G88,3,TRUE) → returns High price

=VLOOKUP(R9,A6:G88,4,TRUE) → returns Low price

The lookup value (R column) was linked to a dropdown-style data entry cell, allowing a user to pick any of the 10+ sampled dates and immediately see the corresponding OHLC values without scrolling through the raw dataset — the same principle behind interactive Excel/BI dashboards.

For the following dataset:	
1	Create a VLOOKUP table and Dropdown menu (together) for the atleast 10 values of dates to k close, shares traded and turnover
2	Prepare any 3 suitable charts for the closing of the June 2021, with all appropriate labelling.

Date	Open	High	Low	Close	Shares Traded	Turnover (Rs. Cr)
1-Mar-21	10304.25	10472.7	10292.75	10411.8	161742834	7265.61
2-Mar-21	10491.8	10766.7	10419.75	10744.05	203848525	9214.92
3-Mar-21	10819.05	10820.4	10640.15	10673.1	145083220	6609.62
4-Mar-21	10559.4	10710.1	10514.3	10625.75	110349226	5527.76
5-Mar-21	10558.15	10663.75	10488.85	10524.4	143548748	6994.24
8-Mar-21	10597.8	10632.1	10488.4	10507.35	129958354	5809.39
9-Mar-21	10572.65	10621.45	10381.6	10464.25	128775369	5034.34
10-Mar-21	10525.75	10613.6	10511.15	10559.3	105320840	4820.76
12-Mar-21	10603.15	10624.35	10313.35	10383.9	93164846	4831.44
15-Mar-21	10446.75	10450.2	10178.25	10335.1	92986219	4192.11
16-Mar-21	10378.3	10401.25	10301.7	10350.8	76199688	3719.98
17-Mar-21	10369.75	10378.8	10133.9	10159.1	101835105	4135.79
18-Mar-21	10251.95	10345.2	10039.55	10143.95	149400391	6107.54
19-Mar-21	10067.15	10207.15	9841.65	10186.5	193169078	7822.19
22-Mar-21	10176.3	10226.8	10069.45	10140.45	97762665	3947.25
23-Mar-21	10183.65	10265	10167.2	10187.65	104759808	4610.4
24-Mar-21	10116.7	10164.35	9897.85	9925.25	109638896	4480.56
25-Mar-21	9916.6	9924.35	9605.05	9651.35	167022338	6347.19
26-Mar-21	9766.4	9853.85	9749.4	9771.5	132881924	5089.31
30-Mar-21	9835.85	9885.95	9749.35	9865	94636811	4524.52
31-Mar-21	9866	9946.95	9812.25	9862.45	98695893	4364.07
1-Apr-21	9967.85	10044.6	9889.7	10022.6	93981628	4167.69
5-Apr-21	9988.1	9991.3	9688.4	9768.05	115412838	4730.76
6-Apr-21	9836.85	9881.2	9755.1	9781.35	98275351	4108.51
7-Apr-21	9790	9950.55	9761.15	9937	83645837	3362.11
8-Apr-21	9992.3	10083.3	9933.1	9959.35	143994891	4864.18
9-Apr-21	9961.55	10032.85	9885.6	9922.95	124222936	4706.09
12-Apr-21	9738.35	9738.35	9345.2	9415.65	171306475	6248.58
13-Apr-21	9438.2	9845.5	9406.4	9818.3	182406912	6980.58
15-Apr-21	9787.85	9817	9558.7	9691.45	124802146	5587.78
16-Apr-21	9691.3	9926.45	9679.5	9808.95	123527904	5262.59
19-Apr-21	9476.65	9562.95	9395.25	9532.15	127131479	4684.37
20-Apr-21	9683.95	9738.1	9539.7	9630.7	115384411	4782.09
22-Apr-21	9529.3	9665.95	9503.8	9644.55	117258633	4926.51
23-Apr-21	9569.25	9695.9	9534.9	9596.95	117321310	4400.76
26-Apr-21	9630.2	9713.1	9612.7	9641.45	69631610	3431.78
27-Apr-21	9662.3	9733.8	9653.9	9708.2	79787691	3908.33
28-Apr-21	9781.5	9968.4	9758.15	9871.5	144645600	7874.9
29-Apr-21	9959.2	9985	9754	9773.7	95398354	4599.68
30-Apr-21	9701.3	9780.35	9619	9640.85	87886826	4426.63

Figure 4: Raw OHLC dataset (left) alongside the dynamic VLOOKUP-driven date lookup table (right).

5.3 Trend Visualization

Three charts were built to analyze the Close price trend for June 2021, including a combination column-and-line chart overlaying daily Turnover (Rs. Cr) against the Close price trend line, making it easy to visually correlate price movement with trading activity.

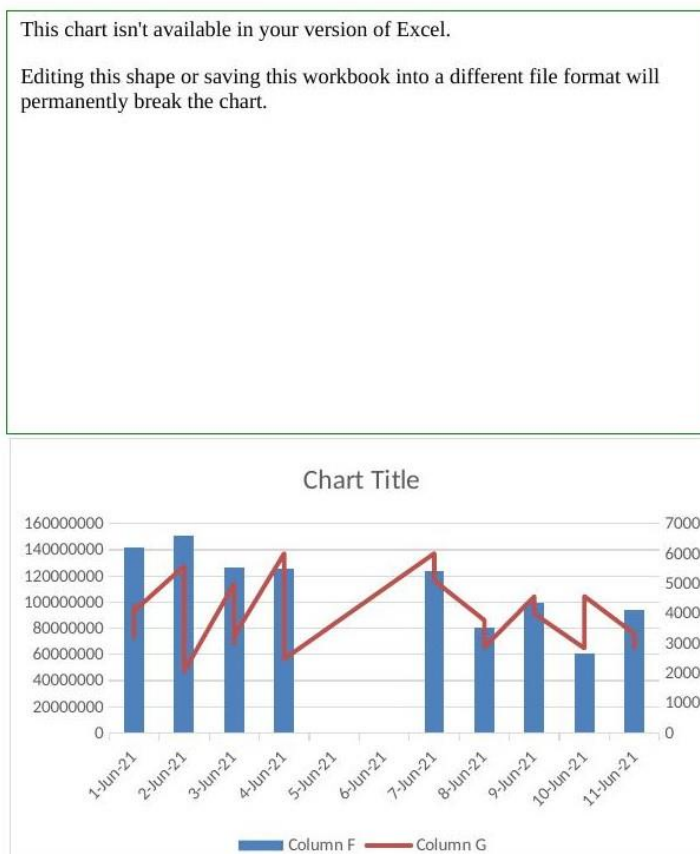


Figure 5: Combination chart — Nifty 50 Close price trend vs. daily Turnover, June 2021.

5.4 Key Insights — Part B

- The Nifty 50 index trended downward from a peak close of ~10,890 in mid-March 2021 to a low of ~9,415 by late March, before staging a broad recovery back above 10,800 by mid-June 2021 — a swing of roughly 15%.
- Days with the highest Turnover generally coincided with the sharpest single-day price moves, visible where the combination chart's column heights spike alongside steep changes in the Close line — consistent with the expected volume-volatility relationship in equity markets.
- The VLOOKUP-based lookup table demonstrates a scalable pattern: the same structure could be extended with more dates, additional metrics, or converted into an XLOOKUP/INDEX-MATCH for a live dashboard.

6. Skills Demonstrated (Relevant to Data Science Roles)

- Data structuring & cleaning — organizing raw tabular data into analysis-ready tables.
- Rule-based classification — equivalent to writing categorical binning logic (e.g., pandas.cut) in Python.
- Group-by aggregation — COUNTIF/SUMIF/AVERAGEIF mirror pandas groupby().agg() operations.
- Ranking & comparative analysis — identifying top/bottom performing segments.
- Lookup-based data retrieval — the same logic used in merge/join operations or dashboard filter controls.
- Data visualization — choosing appropriate chart types (bar for categorical comparison, combination chart for trend + magnitude) and correctly labeling axes and legends.
- Insight communication — translating raw numbers into concise, decision-relevant statements.

7. Conclusion

This case study shows an end-to-end analytical workflow — from raw data to classification, aggregation, lookup automation, and visualization — executed entirely within Excel. It highlights a solid foundation in the analytical thinking that underlies data science tooling more broadly (Python/pandas, SQL, BI platforms), and demonstrates the ability to independently scope a problem, choose the right formula/function, and present findings clearly. This project is included in my portfolio to showcase practical, business-ready data analysis capability.

8. Project Repository

The complete Excel workbook for this case study is available on GitHub:

https://github.com/princemaheta2627-glitch/excel_case_study_2